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Project: CharityFlow — Transparent Charity Donation Platform

Course: MDT915 Blockchain Practitioner

MDT915 — Individual Contribution Report

Specific Contributions

My main role in this project was the documentation and research part. I worked on the written report and the README documentation for the Transparent Charity Donation Platform. My responsibility was to explain how the system works and to organise the technical information in a clear way. I wrote and structured several sections of the documentation. This included the project description the problem statement and the explanation of the solution design. I also documented the system architecture and the data flow of the platform. In this section I explained how the frontend the smart contract and the storage layer interact with each other. Another part of my work was documenting the smart contract functions of the CharityDonation.sol contract. I described the purpose of functions such as createCampaign, donate, withdrawFunds, submitProof, postImpactReport and claimRefund. I also helped organise the table that shows access roles and the description of each function. In addition I worked on the technologies used section where the tools of the project were explained such as Ethereum, Hardhat, Solidity, React, Ethers.js, MetaMask and IPFS through Pinata. I also prepared parts of the user guide testing instructions and the sections about known issues and possible future improvements. My task was to make sure that someone reading the documentation can understand how the platform works and how to run the system locally.

Time Investment

In total I spent around twenty two hours working on the project. Around half of this time was spent reading documentation and learning more about how charity platforms use blockchain for transparency. I also looked at tutorials and examples of Ethereum smart contract systems. The rest of the time was spent writing and editing the report preparing the tables and organising the README documentation. Some time was also spent discussing the system with my teammate, so that the written explanations correctly reflect how the smart contract and the frontend actually work.

Challenges and Solutions

One challenge was understanding the technical structure of the smart contract even though I was not the person implementing the code. The contract includes many functions and different access roles such as donor charity and admin. To solve this I reviewed the code with my teammate who implemented it and asked them to explain the logic of each function. After that I translated the technical behaviour into simpler explanations for the documentation. Another difficulty was the current situation in Dubai which made it harder for us to meet and work together in person. Normally we would have completed many parts of the project together in the same place especially when testing the system and reviewing the report. Instead we had to coordinate online which sometimes slowed down the process and made discussions about technical details longer.

Learning Outcomes

This project helped me understand how blockchain can be used to improve transparency in charity donations. I learned how smart contracts store information about campaigns donations withdrawals and impact reports on the blockchain. I also learned more about how different parts of a decentralised application work together. This includes the frontend interface the smart contract layer and off chain storage such as IPFS. Another important skill I gained was learning how to explain technical systems in written form so that other people can understand how the platform works.

Team Collaboration

Our team divided the work based on different strengths. Since there were only two of us, my teammate focused on smart contract development and frontend implementation. While my focus was the documentation research and report writing. We mainly communicated through online messages and short meetings. We shared updates about our progress and explained the parts we were working on. Overall the collaboration worked well and we completed our parts. One thing that could improve is having more regular meetings earlier in the project. This would make it easier to connect the technical development and the documentation from the beginning and avoid small misunderstandings about how certain parts of the system work.